

BANARAS HINDU UNIVERSITY

B.Sc. (Hons) Semester I Examination, 2024-25 (NEP)

CHEMISTRY**Paper Code: CHEMJ11 : Basic Concepts of Chemistry-I**

Ref. No.: CECNF/N/2024-25/28

Time: 2½ Hours

Maximum Marks: 70

Instructions: Attempt five questions including Question 1 which is compulsory. Attempt at least one question from Questions 2 and 3. The figures in the right-hand margin indicate marks.

Question 1.

- (a) (i*) What is the physical significance of + and - signs in the orbitals? [1]
- (i*) What are the conditions for the acceptability of wave functions obtained by solving the Schrödinger wave equation? [2]
- (b) What are the different consequences of the inert pair effect? [2]
- (c) Obtain the expressions for van der Waals constants a and b in terms of T_c , P_c and R . [3]
- (d) State the first law of thermodynamics and write its limitations. [3]
- (e) Prove that the work done in an isothermal case is higher than in an adiabatic reversible expansion of an ideal gas. [3]

Question 2.

- (a) Calculate the effective nuclear charge on the outermost electron of ^{29}Cu . [2]
- (b) What are eigenfunctions and eigenvalues? Give two examples for each. [2]
- (c) What are the trends of the solubilities of alkali-metal fluorides and iodides in water? Discuss the reasons for the solubility trends of these salts. [4]
- (d) What are the limitations of Born-Landé equation? [2]
- (e) (i*) Calculate the standard reduction potential for Fe^{3+}/Fe . Given $\text{Fe}^{3+}/\text{Fe}^{2+} = +0.77 \text{ V}$ and $\text{Fe}^{2+}/\text{Fe} = -0.47 \text{ V}$. [2]
- (i*) Will Fe^{2+} disproportionate to Fe^{3+} and Fe in an acid solution? Support your answer with reasons. [4]

Question 3.

- (a) Obtain the Schrödinger wave equation assuming the electron behaves like a standing wave in three dimensions. [4]
- (b) Qualitatively discuss the trends of 1st ionization energies for the elements in the 2nd and 3rd rows of the periodic table. [3]

- (c) Discuss the dependence of lattice energy in terms of ionic size and explain the consequences of ionic size on lattice energy and hydration energy. [3]
- (d) Calculate the electronegativity of carbon by Pauling's method if Δ , the extra bond energy for the C-H bond is 24.3 kJ mol^{-1} . Explain, what is meant by extra bond energy. [4]

Question 4.

- (a) Write the basic postulates of kinetic molecular theory of gases giving proper evidences for all of them. [4]
- (b) Explain the terms T_c , P_c and V_c with suitable examples. How will you determine the value of V_c experimentally? [5]
- (c) State the law of corresponding states and derive the expression for van der Waals reduced equation of state. [5]

Question 5.

- (a) The number of drops of liquid and water was observed to be 66 and 30 respectively by the number drop method using a stalagmometer. Calculate the surface tension of the liquid. The surface tension of water is 72.75 dyne/cm and masses of liquid and water are 10.750 g and 11.250 g , respectively for the same volume. [4]
- (b) What is the law of rational indices? What is the need to convert Weiss indices into Miller indices? Write the steps involved. [5]
- (c) Draw and explain the crystal structures of graphite and diamond. [5]

Question 6.

- (a) Using $\mu_{JT} = -\frac{1}{C_p} \left[\left(\frac{dE}{dV} \right)_T \cdot \left(\frac{dV}{dP} \right)_T \right] - \frac{1}{C_p} \left[\frac{d(PV)}{dP} \right]_T$, show under what conditions the Joule-Thomson coefficient will be positive for real gas. [4]
- (b) Why cooling of real gas is observed in Joule-Thomson effect? Show that the Joule-Thomson effect is an isenthalpic process. [5]
- (c) What is the physical significance of entropy? "The entropy of the universe is always increasing." Justify the statement. [5]

Question 7.

- (a) Prove the Maxwell relation $\left(\frac{dP}{dT} \right)_V = \left(\frac{dS}{dV} \right)_T$. [4]
- (b) The total heat absorbed by the system cannot be converted into useful work. Explain the statement based on the Carnot cycle. [5]
- (c) Show that $-(\Delta G)_T = W_{\text{Network}}$. Under what condition the value of Gibbs free energy and work function would be equal? [5]
